DP-FedFace: Privacy-Preserving Facial Recognition in Real Federated Scenarios

Wenjing Wang
Beijing University of Posts and Telecommunications
Haidian Qu, Beijing Shi, China
wwj311@bupt.edu.cn

Si Li
Beijing University of Posts and Telecommunications
Haidian Qu, Beijing Shi, China
lisi@bupt.edu.cn

Abstract

Advanced deep learning-based face recognition models require extensive datasets for optimal performance. However, increasing privacy concerns drive the limitation of face image access on devices to prevent personal information leaks. To address this, federated learning, which allows decentralized data collaboration, has gained popularity. However, traditional federated learning methods risk privacy by transmitting identity proxies to servers. We propose DP-FedFace, a privacy framework specifically designed for a realistic scenario where each client contains only the owner's face images (one identity per client). It uses the difference between human and model perception to eliminate visualization-critical low-frequency components, thus protecting user privacy. We also introduce a novel, learnable privacy cost allocation mechanism that optimizes allocation strategies and adds noise to frequency domain features. Extensive experiments demonstrate that DP-FedFace maintains high recognition accuracy while offers robust privacy protection.

CCS Concepts

• **Security and privacy** → *Privacy protections.*

Keywords

Federated Learning, Privacy-Preserving, Face Recognition, Differential Privacy

ACM Reference Format:
Wenjing Wang and Si Li. 2024. DP-FedFace: Privacy-Preserving Facial Recognition in Real Federated Scenarios. In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM '24), October 21–25, 2024, Boise, ID, USA.* ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3627673.3679901

1 Introduction

Face recognition technology applies broadly across various domains, such as unlocking mobile device, face payment systems, and community access control. Given that these images carry sensitive biometric and identity information, there is an inherent risk of unauthorized data exploitation by service providers and the potential for privacy infringement by malicious entities.

Federated Learning (FL) is a distributed and privacy-aware framework enabling multiple clients to utilize local data for model training, followed by model aggregation and delivery on the server. Clients do not upload their local data to the server or share it with other clients. FedFace [1] pioneers the integration of federated learning into face recognition, addressing the challenge of single-identity client datasets that reflect real-world conditions. It employs average feature initialization for local agents and an enhanced server-side regularizer [18]. However, it requires local devices to transmit identity proxies to the server, contravening federated learning protocol [3]. FedFR [7] aims for personalized face recognition, but its real-world use is constrained by face images typically being uploaded to third-party servers, rather than processed by lightweight clients. Another study [11] leveraging differential privacy for face image protection also overlooks practical application contexts.

To implement federated learning in practical face recognition scenarios, we propose DP-FedFace, a framework addressing the complex issue where each client possesses only a single subject's face images while safeguarding privacy. It extends FedFace by converting client private data into privacy-preserving data, preventing the reconstruction of original images even if identity proxies are uploaded to the server. We leverage block discrete cosine transform to shift images to the frequency domain, enabling the strategic removal of low-frequency components rich in visual information yet less critical for recognition. Furthermore, recognizing the varying importance of different frequency elements for recognition, a learnable privacy cost allocation mechanism dynamically balances privacy and utility, guided by the face recognition network' loss to optimize the allocation of privacy costs. Notably, clients retain only the masked frequency domain features, not the original images, further securing against inadvertent facial information disclosure. Attack simulations confirm our robustness privacy protection measures. Our method achieves a commendable balance between privacy and performance, with significant practical implications. The main contributions of our research include:

- We introduce DP-FedFace, a novel federated learning framework designed for complex and realistic scenarios where each client has face images of a single identity.
- We convert the proprietary face images of clients into the frequency domain, discarding low-frequency components. Additionally, we devise a differential privacy-based learnable cost allocation module. These procedures convert the raw data into privacy-preserving data.
- We perform various experiments, confirming the robust privacy protection of DP-FedFace while demonstrating improved accuracy over pre-trained models and maintaining comparable accuracy to FedFace.

CIKM '24, October 21–25, 2024, Boise, ID, USA

ACM ISBN 979-8-4007-0436-9/24/10
https://doi.org/10.1145/3627673.3679901

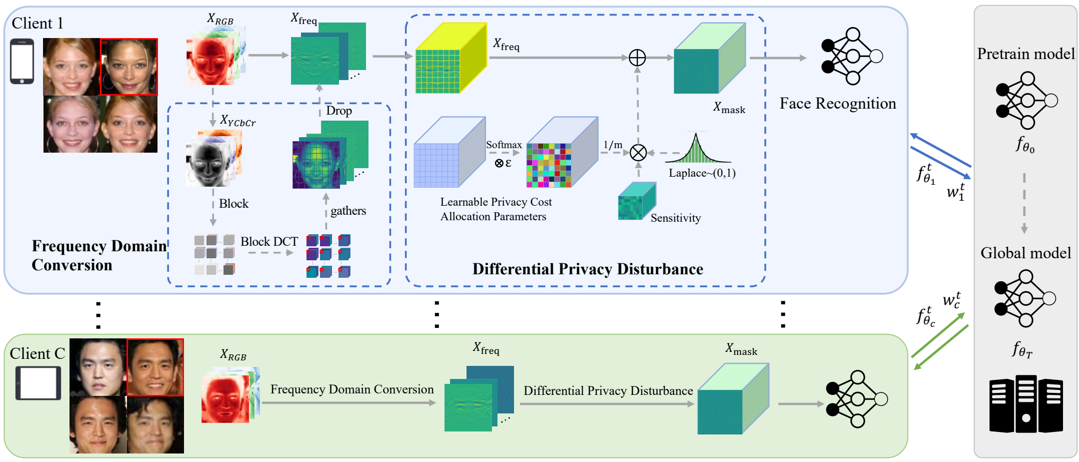

Figure 1: The framework of our proposed DP-FedFace. In the federated learning setup, we work on solving the practical scenario where each client, such as a mobile phone or iPad, corresponds to a single user.

2 Method

2.1 Overview of DP-FedFace

Figure 1 illustrates the architecture of DP-FedFace. Instead of starting from scratch, we initially train the face recognition model on a publicly accessible dataset to achieve superior results.

In our federated learning framework, we consider a realistic scenario with C local client nodes, each possessing face images of a single individual, and one central server with a pre-trained face feature extractor f_{θ_0}. Each local client i is initialized with $f_{\theta_i} = f_{\theta_0}$.

Assuming client i has a dataset $D_i = \left\{(x_i^1, y_i), (x_i^2, y_i), \dots (x_i^{n_i}, y_i)\right\}$, the initialized class embedding can be represented as:

$$w_i^0 = \frac{1}{n_i}\sum_{j=1}^{n_i} f_{\theta_0}(x_i^j). \tag{1}$$

For the derived w_i^t in the t-th communication round, we apply normalization, thereby confining the class embedding to a d-dimensional hypersphere.

The local client comprises three modules: the frequency domain conversion module, the differential privacy disturbance module, and the face recognition module. Initially, the frequency domain conversion module transforms the original input images into frequency domain. Subsequently, the differential privacy disturbance module generates learnable noise for incorporation into the frequency domain features. Finally, these noisy features serve as input for the training of face recognition network.

Given that the input features becomes C channels, we adjust the initial convolutional layer's input channel of the face recognition network from 3 to C. Since each client has only one ground truth, we adopt squared hinge loss with cosine distance for back-propagation to learn network parameters and privacy cost allocation parameters. The training objective is as follows:

$$l(f_\theta^t(x,i)) = \max(0, m - (w_i^t)^T f_\theta^t(x))^2. \tag{2}$$

Clients transmit the parameters of the feature extractor θ and class embeddings w_i back to the central server. The server then utilizes an algorithm akin to FedAvg [10] to execute a weighted average aggregation of the uploaded parameter values as follows:

$$\theta^{t+1} = \sum_{i=1}^{C}\frac{n_i}{N}\theta_i^t. \tag{3}$$

Given that each client solely communicates its unique class embedding w_i, the server concatenates the class embeddings from all clients to formulate the comprehensive class embedding matrix by:

$$W^t = [w_1^t, w_2^t, \cdots, w_c^t]^T. \tag{4}$$

To prevent performance degradation from overlapping class embeddings, we use a spreadout regularizer [18], ensuring distinct separation within the feature space:

$$reg_{sp}(W^t) = \sum_{c=1}^{C}\sum_{\hat{c}\neq c}(\max\left\{0, v - d(w_c^t, w_{\hat{c}}^t)\right\})^2, \tag{5}$$

where v stands for the margin.

The updated global model parameters θ^t and class embeddings w_i^t are distributed to clients. This process repeats for T rounds until convergence, yielding the global model parameters θ^T of the face feature extractor.

2.2 Frequency Domain Conversion module

Starting from a unique viewpoint, we explore the contrasting perceptions of images between human and machine models. Recent studies in image classification reveal that machine model predictions predominantly depend on the high-frequency components of images, which are virtually imperceptible and largely unrecognizable to humans [6, 14, 16]. Therefore, our methodology works on eliminating the low-frequency channels discernible to humans and capitalizing on the remaining high-frequency channels.

Inspired by the JPEG compression methodology, we employ the Block Discrete Cosine Transform (BDCT) to facilitate the transition from the image domain to the frequency domain. This transition is notable for its simplicity, rapidity, and ease of deployment. The procedure begins with an eight-fold upsampling of each input image, employing bilinear interpolation to maintain the image size constant for subsequent feeding into the face recognition network.

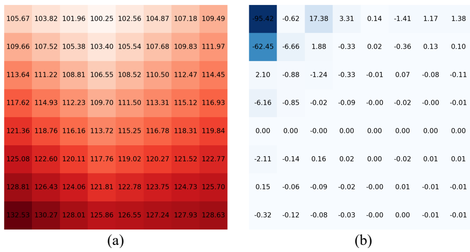

Figure 2: (a) One block of the original image with size 8×8. (b) The frequency-domain features of the image block after DCT transformation.

Following this, we convert the images from RGB to YCbCr color space. To satisfy the requirements of the BDCT input, we normalize the image values to $[-128, 127]$. Subsequently, we partition the image into $\frac{H}{8} \times \frac{H}{8}$ blocks, each of which measures 8×8.

The next step involves applying a normalized 2D type-II DCT to each 8×8 block, converting spatial information into frequency information. This process restructures the original $[3, H, W]$ image into a $[3, 64, H, W]$ format, aggregating elements of identical positions within each block into the same channel.

In Figure 2, we visualize an 8×8 block. The results show that upon its transition into the frequency domain, the more substantial absolute values are primarily concentrated in the upper left region. Low-frequency channels with high energy are essential for visualization, yet they exert limited influence on model recognition. Consequently, before advancing to the subsequent modules, we remove $\lambda_1 = 3$ channels characterized by high energy, randomly discard $\lambda_2 = 10$ remaining channels, and preserve the rest.

2.3 Differential Privacy Disturbance Module

Differential Privacy (DP) [4] is a framework under which very little about any specific individual can be learned, regardless of the attacker's prior knowledge or information source. For two datasets, they are called neighboring datasets if only one individual is different between them. For every pair of neighboring datasets X, X' and every possible output T, an algorithm M is ϵ-DP if

$$Pr(M(X) = T) \leq e^{\epsilon} \cdot Pr(M(X') = T), \tag{6}$$

where ϵ represents the privacy cost. Optimal privacy protection is achieved when $\epsilon = 0$.

In the differential privacy disturbance module, we add noise to the frequency domain features that have passed through the frequency domain conversion module, with the objective of bolstering privacy protection. This necessitates adapting the initial database domain differential privacy approach to the frequency domain.

We anticipate that any pair of BDCT representations will keep distances between similar images small and those between dissimilar images noticeably larger. After frequency domain conversion, the image is transformed into a frequency domain representation with size $[C, H, W]$. $R_{i,j,k} = [r_{min}^{i,j,k}, r_{max}^{i,j,k}]$ is defined as the sensitivity of the element situated at the $[i, j, k]$ position within the feature of the frequency domain. For two elements x and x' in the BDCT representation, we establish the distance between them as follows:

$$d_{i,j,k}(x, x') = \frac{|x - x'|}{r_{max}^{i,j,k} - r_{min}^{i,j,k}}, \quad \forall x, x' \in R_{i,j,k}. \tag{7}$$

Therefore, the distance between the entire frequency domain representations is defined below:

$$d(X, X') = \max(d_{i,j,k}(x, x')), \quad \forall X, X' \in R^{H,W,C}. \tag{8}$$

Thus, considering any representation X, X' and potential output $\Phi \in \mathcal{R}^{H,W,C}$, we can assert that an algorithm M adheres to ϵ-DP if

$$Pr(M(X) = \Phi) \leq e^{\epsilon d(X,X')} \cdot Pr(M(X') = \Phi). \tag{9}$$

Following the theoretical analysis of differential privacy within the frequency domain representation, we can derive the subsequent findings: The BDCT representation of any image $X \in \mathcal{R}^{H,W,C}$ can be protected by ϵ-DP through the addition of a vector $Y \in \mathcal{R}^{H,W,C}$. Here, each variable $Y_{i,j,k}$ independently follows a Laplace distribution with a specified scaling parameter

$$\sigma_{i,j,k} = \frac{r_{max}^{i,j,k} - r_{min}^{i,j,k}}{\epsilon_{i,j,k}}, \quad \text{where} \sum_{i,j,k} \epsilon_{i,j,k} = \epsilon. \tag{10}$$

The frequency domain conversion module produces an output with size $[H, W, C]$. To mask the frequency characteristics, a noise matrix of the same size is essential. Given DCT properties, the significance of frequency attributes for face recognition is not uniform. To optimize accuracy and privacy, we have incorporated a learnable privacy cost allocation to dedicate a more significant portion of the privacy cost to areas that are crucial for face recognition.

Leveraging prior insights, we employ a Laplace transform with a scaling parameter of $\sigma_{i,j,k} = \frac{r_{max}^{i,j,k} - r_{min}^{i,j,k}}{\epsilon_{i,j,k}}$, to ensure ϵ-DP protection for frequency domain features. We convert the training set to the frequency domain and calculate the maximum and minimum values at each position, with sensitivity being defined as the difference between these two values. We then sample from the Laplace distribution based on the sensitivity and the inverse of privacy cost. The resulting noise is added one-to-one to the output features of the frequency domain conversion module and transmitted to the face recognition network. The privacy cost allocation parameters are optimized through the loss function of the face recognition model.

3 Experiment

3.1 Experimental Setup

Datasets. The CASIA-WebFace [17] contains 494,414 images from 10,575 different subjects. To simulate a real face recognition federated learning scenario, we randomly select 10,000 subjects from the CASIA-WebFace dataset as the training set, with 6,000 identities used for pre-training the face recognition model, and the remaining 4,000 identities evenly distributed among 4,000 clients. We conduct extensive experiments on several popular face recognition benchmarks, including five widely used standard-sized test datasets, LFW [5], CFP-FP [13], CPLFW [19], CALFW [19], and AgeDB [12], as well as two large-scale datasets, IJB-B [15] and IJB-C [9].

Implementation Details. During data preprocessing stage, we use five facial landmarks for similarity transformation, resize the face images to 112×112, and normalize them. We utilize the same 64-layer CNN architecture from [8] as the backbone, and modify the input channel of the initial convolutional layer from 3 to 153. For the pre-training of the global face recognition model on the

Method (%)	Distributed	LFW	CFP-FP	AgeDB-30	CALFW	CPLFW	IJB-B(1e-3)	IJB-C(1e-3)
Baseline (6000IDs)	No	98.27	88.99	87.70	90.33	97.91	80.38	83.25
Fine-tuning baseline (10000IDs)	No	98.82	90.87	90.42	91.88	98.77	84.48	87.75
FedFace [1]	Yes	99.01	90.23	90.38	91.88	98.62	83.79	88.21
Proposed DP-FedFace	Yes	98.48	89.37	90.13	90.95	98.54	83.42	87.76

Table 1: Comparison of the face recognition accuracy among various benchmarks.

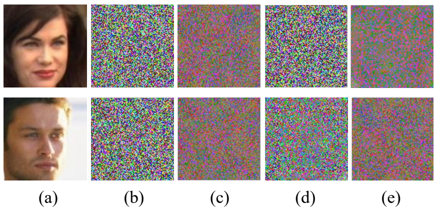

Figure 3: Visualisation of white-box attacks. (a) Raw image. (b) IDCT with $\epsilon = 0.5$. (c) Denoising image of (b). (d) IDCT with $\epsilon = 100$. (e) Denoising image of (d).

Method	LFW	CFP-FP	AgeDB-30
w low frequency	51.529	51.307	52.464
w/o low frequency	20.512	20.546	21.010

Table 2: PSNRs with or without low-frequency components.

Method(%)	LFW	CFP-FP	AgeDB-30
Fixed	75.96	72.16	70.28
Learnable	98.52	90.39	90.73

Table 3: Comparison of the face recognition accuracy among different privacy cost allocation methods.

server, we apply an angular margin softmax loss with a scale hyper-parameter value of 30 and a margin hyper-parameter value of 10. In the Federated Learning setup at each individual client, we employ a squared hinge loss with cosine distance and a margin of 0.9 as the positive loss function. We normalize both the instance embeddings and the class embeddings to ensure their norms equal 1. The model is trained for 20 communication rounds. To compare fairly with FedFace, we keep the learning rate at 0.01, using the SGD optimizer with the momentum to 0.9, and the weight decay to 5e-4.

3.2 Benchmarks on recognition accuracy

We compare DP-FedFace algorithm with the following algorithms:

Baseline: This involves pre-training model with face images of the 6,000 identities from the CASIA-WebFace dataset. Additionally, this model is used to initialize the server-side global model.

Fine-tuning baseline: This approach directly aggregate face images from 4,000 clients to enhance the pre-trained model in a non-federated manner. This is the traditional training method for face recognition models based on deep neural networks.

FedFace [1]: FedFace improves the pre-trained face recognition systems by collaboratively learning from client data. However, when the parameters of the feature extractor are uploaded to the server, the class embedding matrix is simultaneously uploaded, posing a serious privacy disclosure risk.

Table 1 presents the verification performances on various benchmarks. Across all datasets, particularly the two large-scale ones, FedFace consistently exhibits a significant advantage over the baseline. This suggests that federated learning can effectively leverage distributed client data to improve the performance of the global model. Furthermore, DP-FedFace not only displays an enhancement in accuracy compared to the baseline, implying that the model's ability to continually enhance its performance is unaffected even with the privacy protection settings, but also maintains an accuracy level comparable to FedFace. This provides a compelling demonstration of our approach's superiority.

3.3 White-box Attacking Experiments

Assuming a white-box attack, the attacker is presumed to have knowledge of all operations on local private face images. Consequently, he may attempt to reverse engineer the frequency domain features by performing an inverse discrete cosine transform (IDCT) on the transmitted data. Since the operation of deleting certain frequency components is irreversible, these components are entirely filled with zeros. Subsequently, we utilize the non-local means denoising [2] method for noise reduction. The effectiveness of privacy protection decreases as the privacy cost increases. Figure 3 shows that even with a high privacy cost of 100, the recovered images are replete with noise, resulting in an inability to extract any visual information about the user's face.

3.4 Ablation Study

Effects of removing low-frequency components. To show the impact of removing low-frequency components, we compare models with and without these components. A high privacy cost, fixed at 20, is set to differentiate the two scenarios. As shown in Table 2, the substantial PSNR decrease indicates that removing low-frequency significantly boosts information protection.

Effects of learnable privacy cost. We compare the accuracy of learnable and fixed privacy costs, with the total privacy cost of all elements set to $\epsilon = 4$. The pre-trained model includes the frequency domain conversion module, which uses noise-free frequency domain features as inputs for the face recognition model. As discernible from Table 3, the use of fixed privacy cost is markedly inferior to that of learnable privacy cost.

4 Conclusion

Our novel DP-FedFace is a privacy-conscious federated learning method for face recognition. It is designed to address real-world scenarios where a single client possesses face images of only one single identity, ensuring robust privacy safeguards and maintaining the face recognition accuracy. By applying differential privacy to inject noise into the frequency domain transformed images, clients achieve robust privacy protection. This ensures that attackers cannot recover the face images from the uploaded class embeddings.

References

[1] Divyansh Aggarwal, Jiayu Zhou, and Anil K Jain. 2021. Fedface: Collaborative learning of face recognition model. In *2021 IEEE International Joint Conference on Biometrics (IJCB)*. IEEE, 1–8.

[2] Antoni Buades, Bartomeu Coll, and J-M Morel. 2005. A non-local algorithm for image denoising. In *2005 IEEE computer society conference on computer vision and pattern recognition (CVPR'05)*, Vol. 2. Ieee, 60–65.

[3] Chi Nhan Duong, Thanh-Dat Truong, Khoa Luu, Kha Gia Quach, Hung Bui, and Kaushik Roy. 2020. Vec2face: Unveil human faces from their blackbox features in face recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 6132–6141.

[4] Cynthia Dwork, Frank McSherry, Kobbi Nissim, and Adam Smith. 2006. Calibrating noise to sensitivity in private data analysis. In *Theory of Cryptography: Third Theory of Cryptography Conference, TCC 2006, New York, NY, USA, March 4-7, 2006. Proceedings 3*. Springer, 265–284.

[5] Gary B Huang, Marwan Mattar, Tamara Berg, and Eric Learned-Miller. 2008. Labeled faces in the wild: A database forstudying face recognition in unconstrained environments. In *Workshop on faces in'Real-Life'Images: detection, alignment, and recognition*.

[6] Jiazhen Ji, Huan Wang, Yuge Huang, Jiaxiang Wu, Xingkun Xu, Shouhong Ding, ShengChuan Zhang, Liujuan Cao, and Rongrong Ji. 2022. Privacy-preserving face recognition with learnable privacy budgets in frequency domain. In *European Conference on Computer Vision*. Springer, 475–491.

[7] Chih-Ting Liu, Chien-Yi Wang, Shao-Yi Chien, and Shang-Hong Lai. 2022. FedFR: Joint optimization federated framework for generic and personalized face recognition. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 1656–1664.

[8] Weiyang Liu, Yandong Wen, Zhiding Yu, Ming Li, Bhiksha Raj, and Le Song. 2017. Sphereface: Deep hypersphere embedding for face recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 212–220.

[9] Brianna Maze, Jocelyn Adams, James A Duncan, Nathan Kalka, Tim Miller, Charles Otto, Anil K Jain, W Tyler Niggel, Janet Anderson, Jordan Cheney, et al. 2018. Iarpa janus benchmark-c: Face dataset and protocol. In *2018 international conference on biometrics (ICB)*. IEEE, 158–165.

[10] Brendan McMahan, Eider Moore, Daniel Ramage, Seth Hampson, and Blaise Aguera y Arcas. 2017. Communication-efficient learning of deep networks from decentralized data. In *Artificial intelligence and statistics*. PMLR, 1273–1282.

[11] Qiang Meng, Feng Zhou, Hainan Ren, Tianshu Feng, Guochao Liu, and Yuanqing Lin. 2022. Improving federated learning face recognition via privacy-agnostic clusters. *arXiv preprint arXiv:2201.12467* (2022).

[12] Stylianos Moschoglou, Athanasios Papaioannou, Christos Sagonas, Jiankang Deng, Irene Kotsia, and Stefanos Zafeiriou. 2017. Agedb: the first manually collected, in-the-wild age database. In *proceedings of the IEEE conference on computer vision and pattern recognition workshops*. 51–59.

[13] Soumyadip Sengupta, Jun-Cheng Chen, Carlos Castillo, Vishal M Patel, Rama Chellappa, and David W Jacobs. 2016. Frontal to profile face verification in the wild. In *2016 IEEE winter conference on applications of computer vision (WACV)*. IEEE, 1–9.

[14] Haohan Wang, Xindi Wu, Zeyi Huang, and Eric P Xing. 2020. High-frequency component helps explain the generalization of convolutional neural networks. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 8684–8694.

[15] Cameron Whitelam, Emma Taborsky, Austin Blanton, Brianna Maze, Jocelyn Adams, Tim Miller, Nathan Kalka, Anil K Jain, James A Duncan, Kristen Allen, et al. 2017. Iarpa janus benchmark-b face dataset. In *proceedings of the IEEE conference on computer vision and pattern recognition workshops*. 90–98.

[16] Kai Xu, Minghai Qin, Fei Sun, Yuhao Wang, Yen-Kuang Chen, and Fengbo Ren. 2020. Learning in the frequency domain. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 1740–1749.

[17] Dong Yi, Zhen Lei, Shengcai Liao, and Stan Z Li. 2014. Learning face representation from scratch. *arXiv preprint arXiv:1411.7923* (2014).

[18] FX Yu, AS Rawat, AK Menon, and S Kumar. [n. d.]. Federated Learning with Only Positive Labels. arXiv 2020. *arXiv preprint arXiv:2004.10342* ([n. d.]).

[19] Tianyue Zheng, Weihong Deng, and Jiani Hu. 2017. Cross-age lfw: A database for studying cross-age face recognition in unconstrained environments. *arXiv preprint arXiv:1708.08197* (2017).